

Ventilated Brake Disk Air Streamlining Using Curved Vane

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1 Introduction

This research highlights the potential of curved vane designs to significantly enhance the performance of ventilated brake disks, contributing to better vehicle safety and braking efficiency. Traditional ventilated brake disks, which use straight radial blades, suffer from flow misalignment, leading to inefficient heat dissipation. The authors propose two innovative curved blade designs aimed at enhancing airflow through the brake disk's ventilated channels.

Using Computational Fluid Dynamics simulations in Ansys Fluent, the study analyses and compares the traditional design with the new curved designs. The CFD results show a significant improvement in mass flow rate for the curved designs, with Curved 1 showing a 51% increase in airflow, and Curved 2 showing a 32% increase compared to the baseline confirming that the curved blade designs reduce flow recirculation and improve overall cooling efficiency [1].

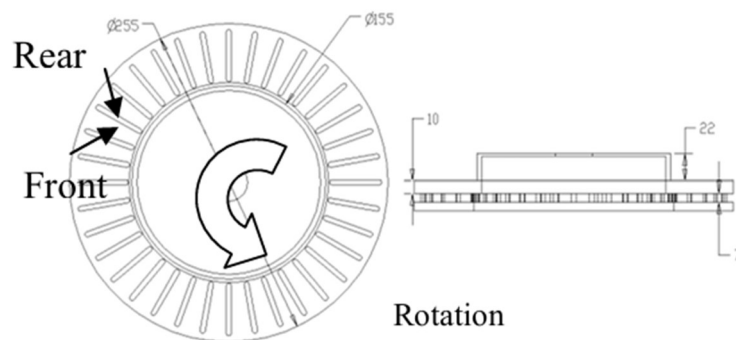


Figure 3: Traditional brake disk

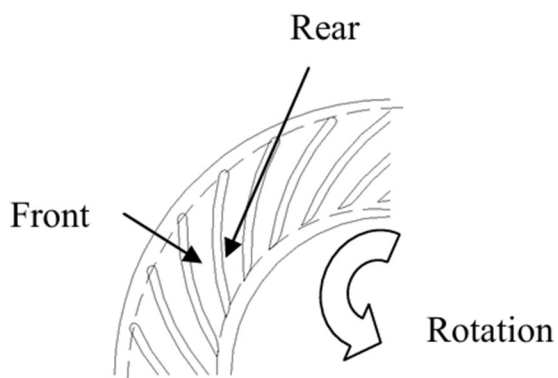


Figure 1: Curved 1 ventilated disk brake design

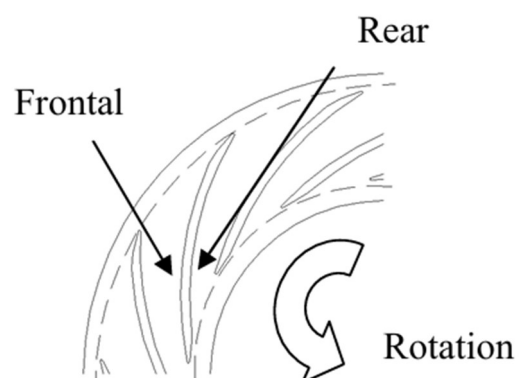


Figure 2: Curved 2 ventilated disk brake design

2 Assumptions and Simplifications

Nature of the problem:

In this experiment, they had considered about three vanes near each other as the flow domain. The flow is internal, three dimensional. Since in this problem the properties of the problem, such as velocities and pressure, do not vary with time, the system is assumed to be in steady state condition.

Geometry:

The baseline design is a cast iron ventilated brake disc from a commercial vehicle. The brake disk has 36 equally spaced radial vane or blades. It is assumed that the flow is turbulent across the entire range of rotational speeds.

Table 1 : Characteristics of baseline brake disk [1].

Outer diameter	255mm
Inner diameter	145mm
Plate thickness	7mm
Blade passage height	10mm
Number of blades	36

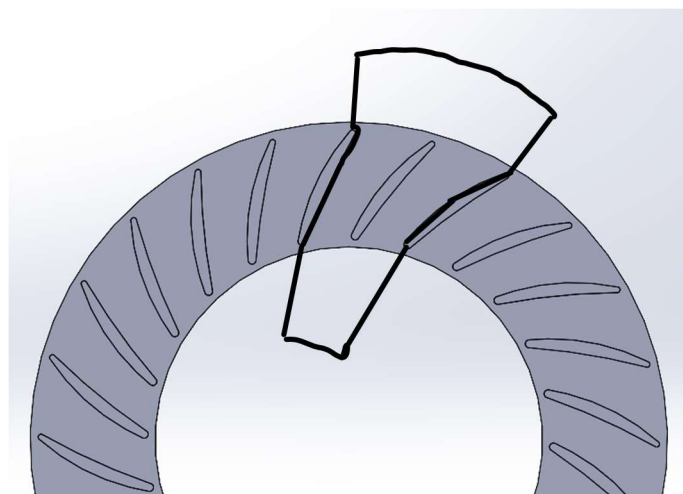


Figure 4: Control volume of the experiment

With the rotation of the wheel, the air is entering into the domain with around 45 degrees angle to the horizontal direction. To capture this, the domain is changed. The air is entering around 26ms^{-1} . At the outer side, the air is entering to the domain from the left side vanes. It also affects to the air flow. Due to the rotation air is entering to the domain around 24ms^{-1} from the outside.

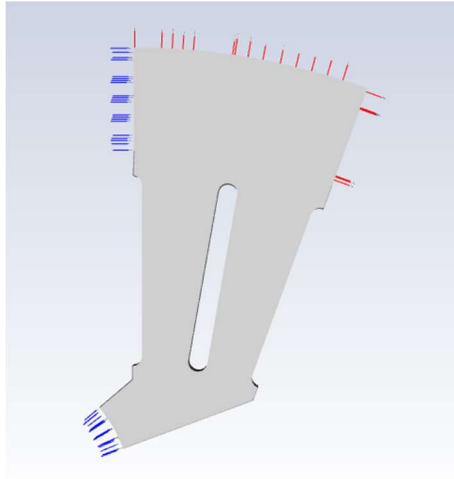


Figure 5: Flow domain with two inlets and outlets

Properties of the fluid:

- Air is used for the experiment. Assume the temperature is constant throughout the process.
- The fluid is incompressible.
- Density: 1.225 kgm^{-3}
- Viscosity: $1.7894 \times 10^{-5}\text{ kgm}^{-1}\text{s}^{-1}$

3 Flow Domain and Computational Mesh

3.1 Flow domain

The purpose of this experiment is to improve the mass flow rate through the geometry modification of the brake disc. As a result, three different geometries were developed. For each of these geometries, separate flow domains were designed. In the literature, the dimensions of the flow domains are not specified.

In the literature, the experiment described was performed on 3 adjacent vanes. The flow domain was divided into 12 equal sections with one section used for the simulation, given that the brake disc contains 36 vanes. A definition of the internal flow domain was provided based on the dimensions of the brake disc specified in literature. The dimensions of the baseline design were referenced from the literature, and the vane designs for the other two curved configurations were developed from Figure 2 and 3 [1].

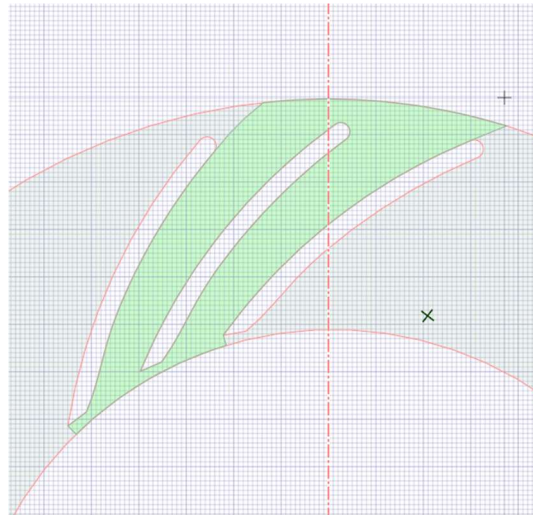


Figure 6: Top view of the curve 1 internal flow domain

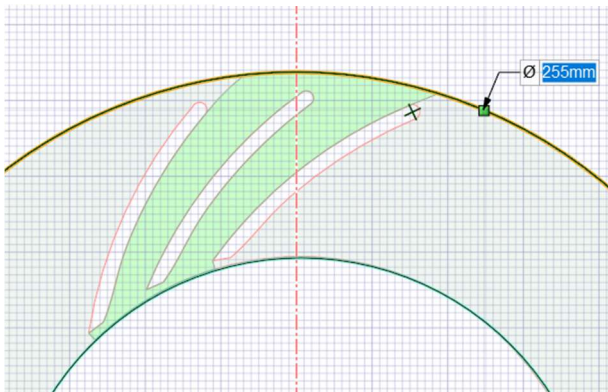


Figure 8: Outer diameter of the brake disk

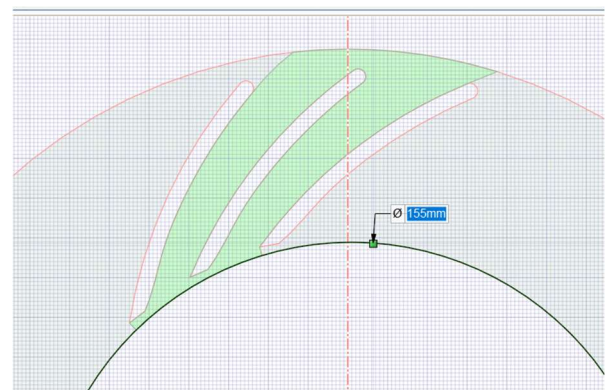


Figure 7: Inner diameter of the brake disk

Two boundaries at outside of the disk and the inside of the disk are defined to capture the external flow of the air.

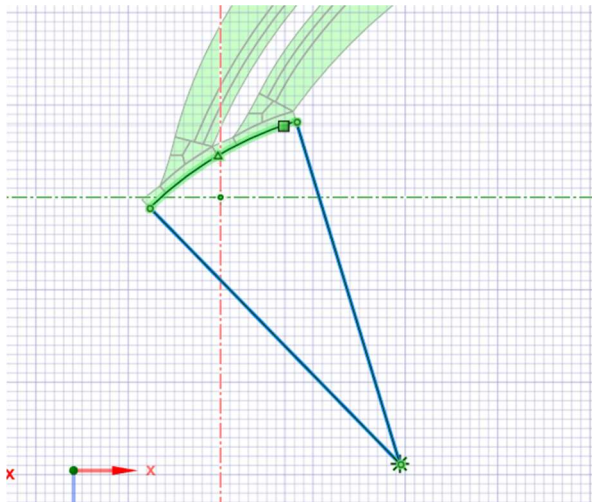


Figure 9: Inside boundary of the inside diameter

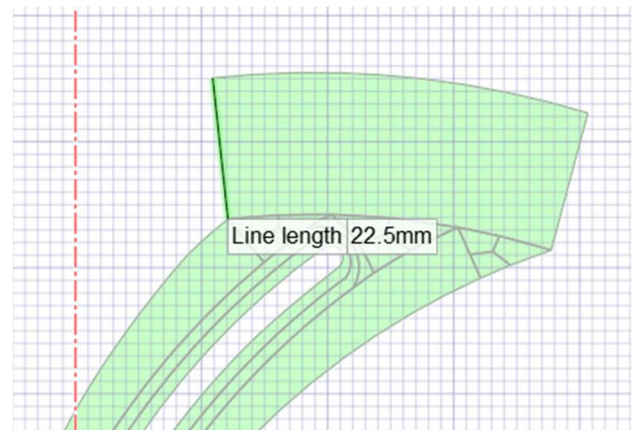


Figure 10: Outside boundary of the outside diameter

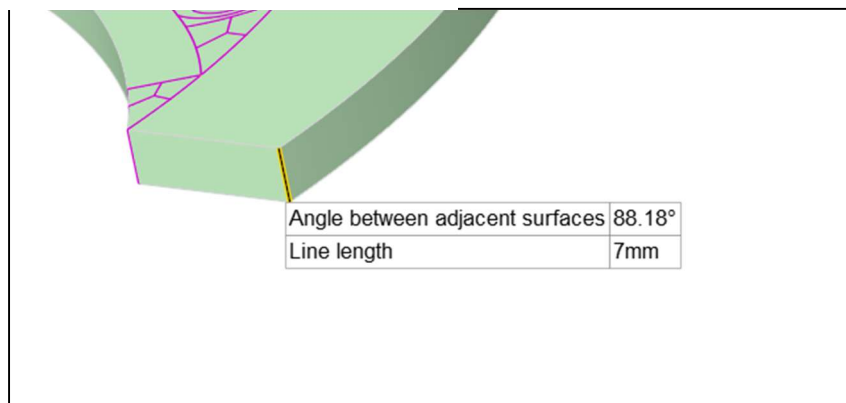


Figure 11: Height of the domain

The final flow domains for the three different geometries are given below.

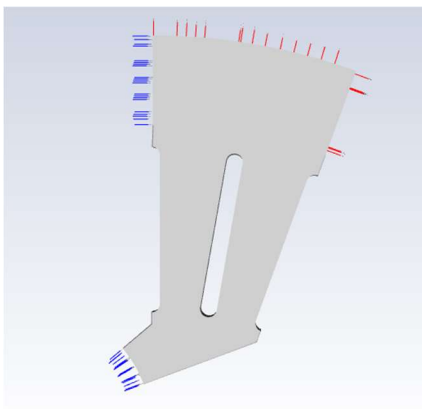


Figure 13: flow domain
for curve baseline

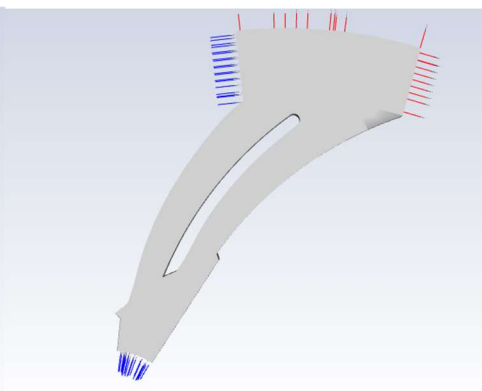


Figure 14: flow domain for
geometry 1

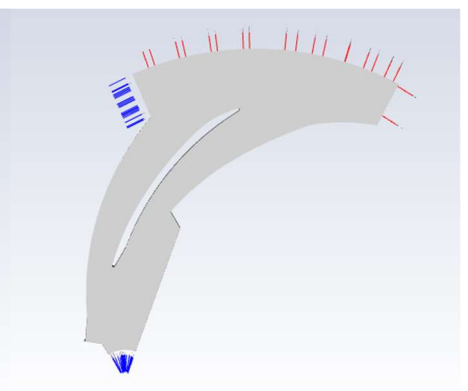


Figure 12: flow domain for
geometry 2

3.2 Mesh

According to the literature, the computational model was constructed in GAMBIT for the two flow channels consist of about 2.5 million hexahedral, tetrahedral, and pyramid cells. The boundary layer near the wall is captured using cell height of 0.1mm and followed by inflation ratio of 1.2[1].

Due to the undefined boundaries and the mesh in the literature, the mesh is designed as suitable for the geometry.

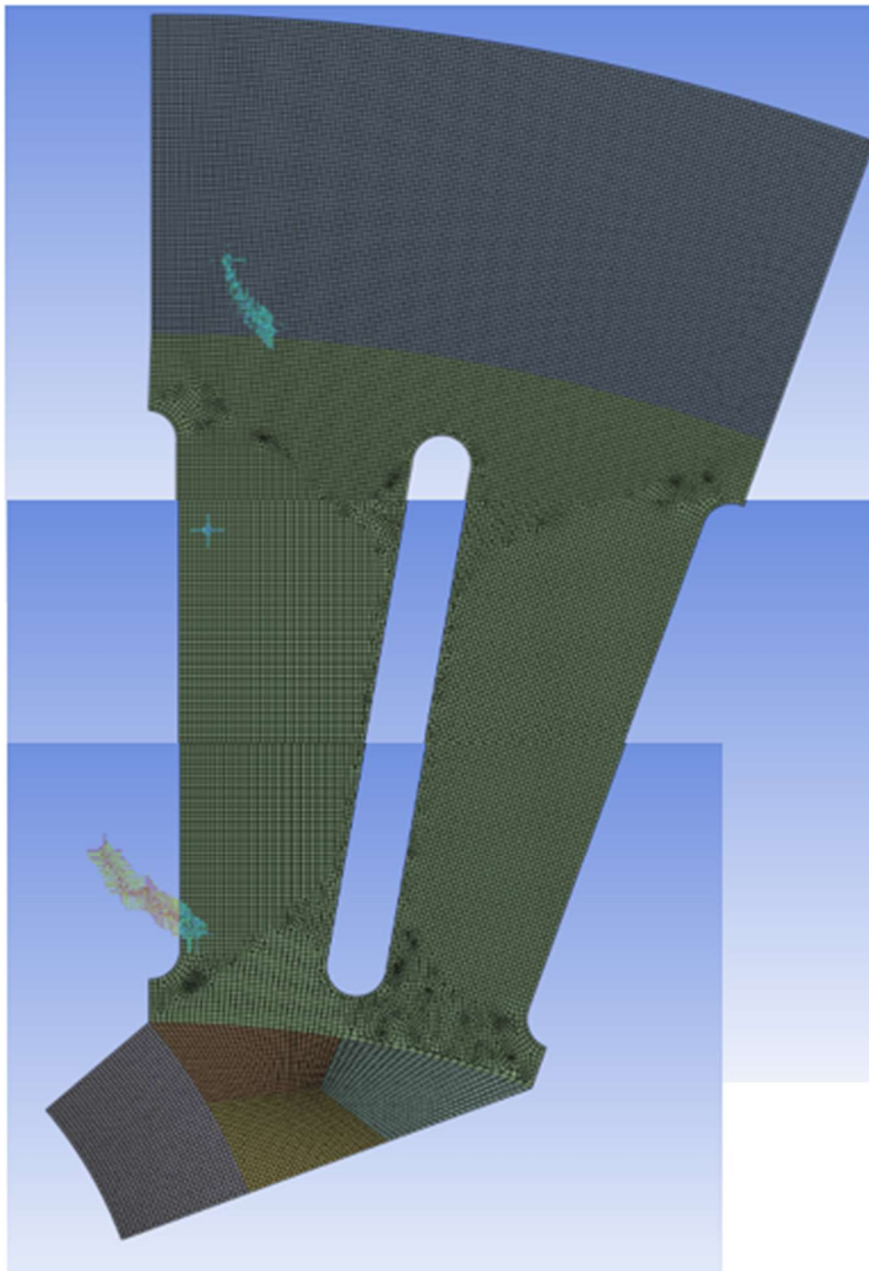


Figure 15: Mesh for the baseline

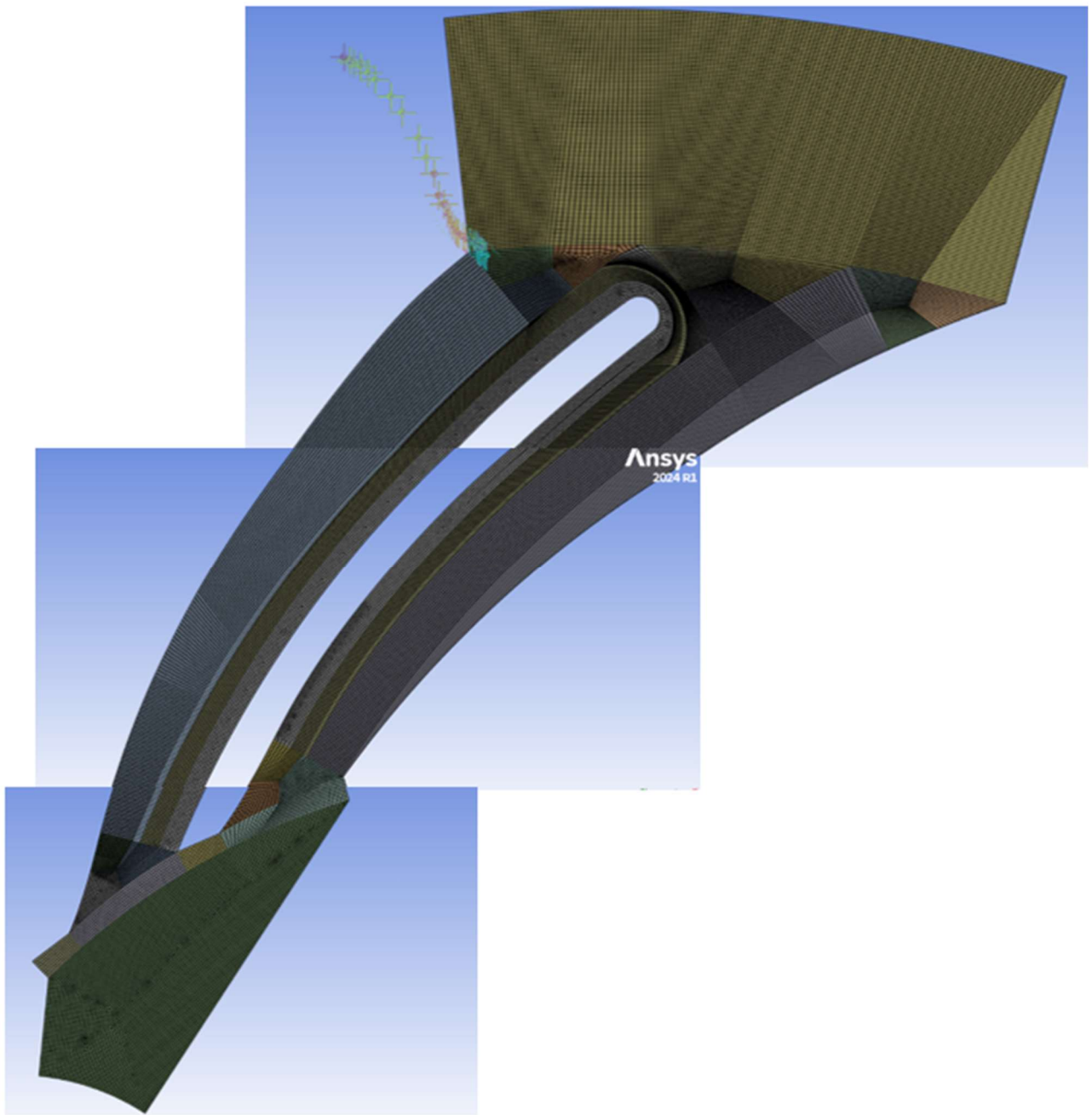


Figure 16: Mesh for the curve 1



Figure 17: Mesh for the curve 2

3.3 Mesh Quality

Table 2: Mesh Quality average values

Mesh matrices	Baseline	Curve 1	Curve 2
Element quality (Average)	1.2797	2117350	0.94534
Aspect ratio (Average)	1.2797	1.5101	1.2797
Skewness (Average)	3.7708×10^{-2}	0.12691	1.2797×10^{-2}
Orthogonal Quality (Average)	3.7708	0.95327	1.2797
Elements	748213	1350552	2117350

Mesh matrices for baseline

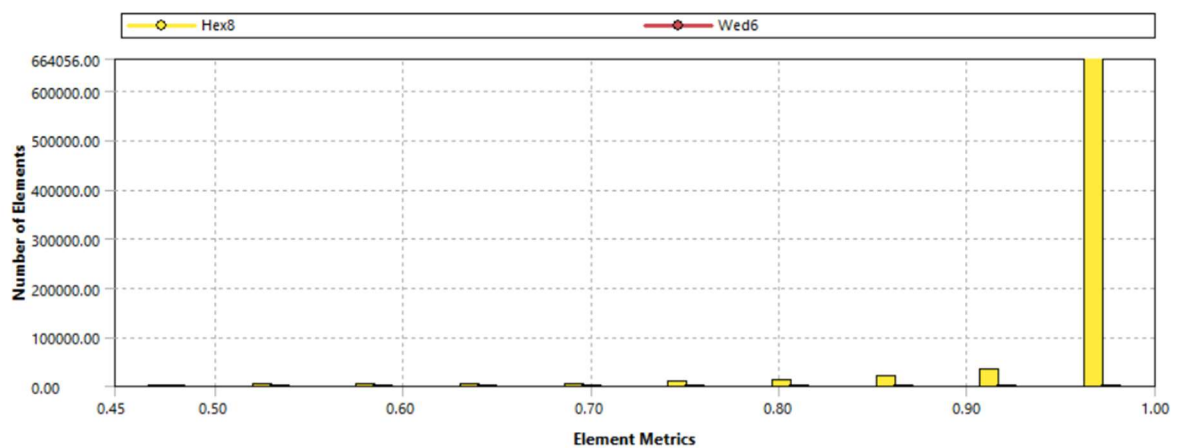


Figure 18: Element quality for baseline

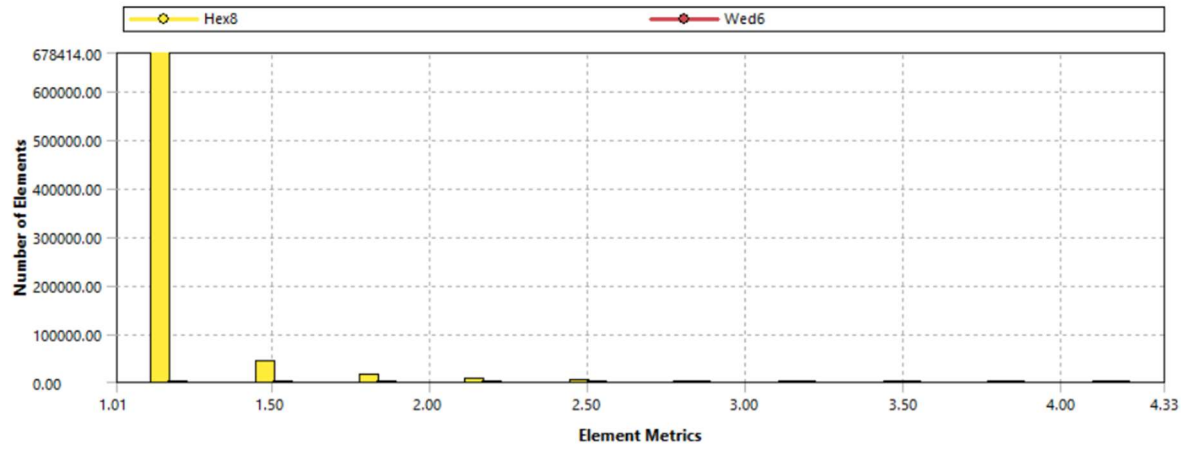


Figure 19: Aspect ratio for baseline

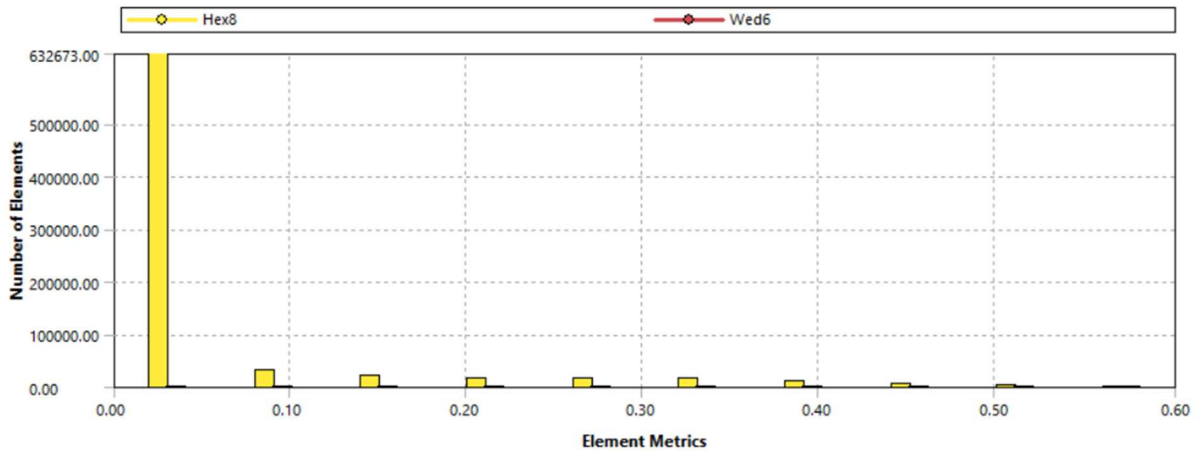


Figure 20: Skewness for baseline

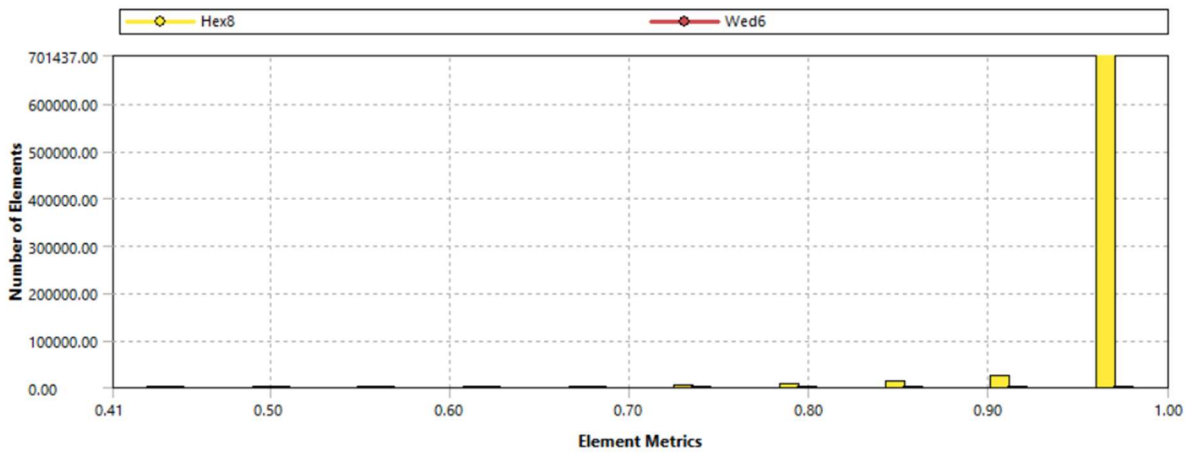


Figure 21: Orthogonal quality for baseline

Mesh matrices for curve 1

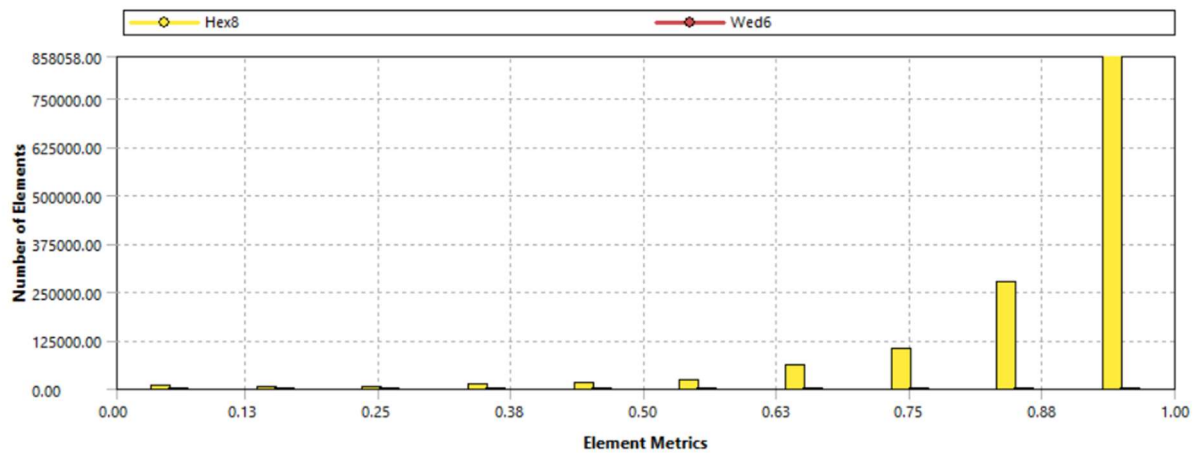


Figure 22: Element quality for curve 1

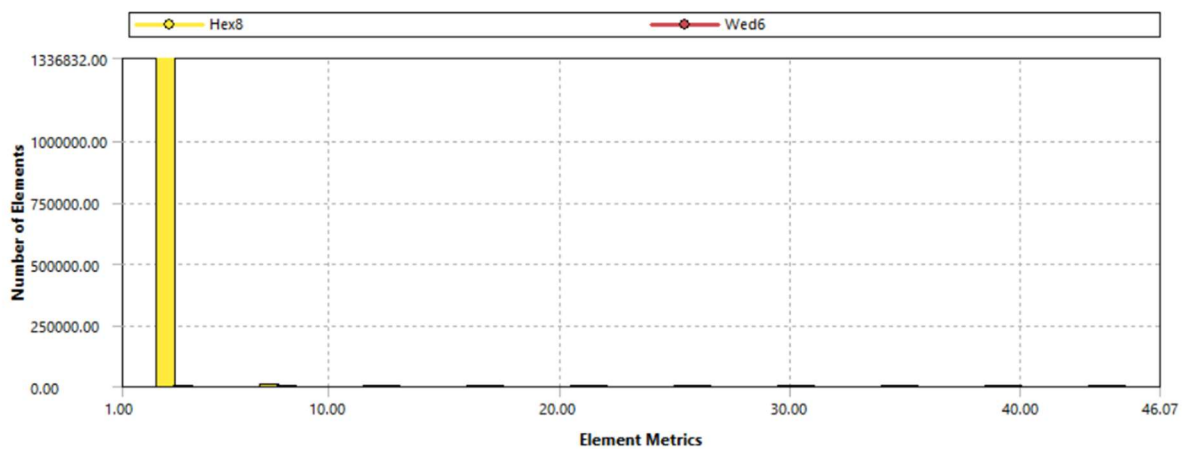


Figure 23: Aspect ratio for curve 1

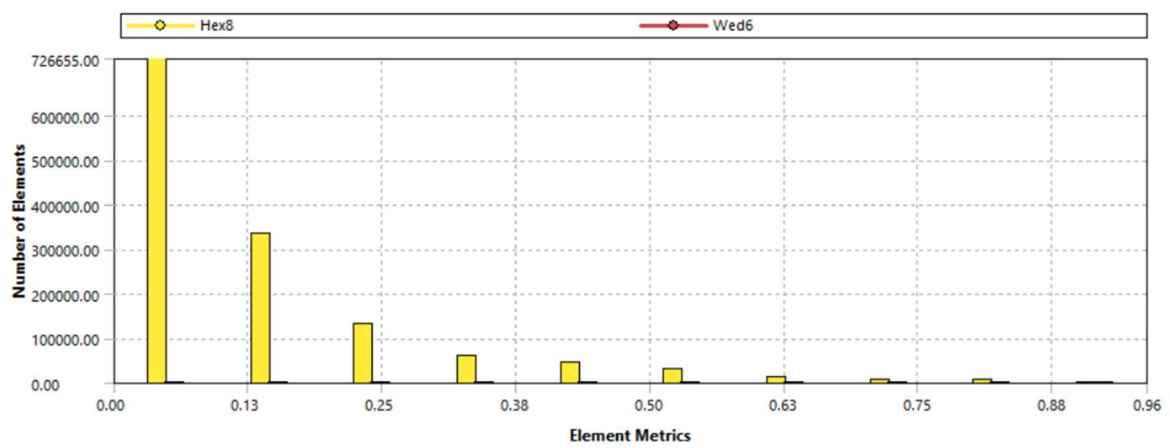


Figure 24: Skewness for curve 1

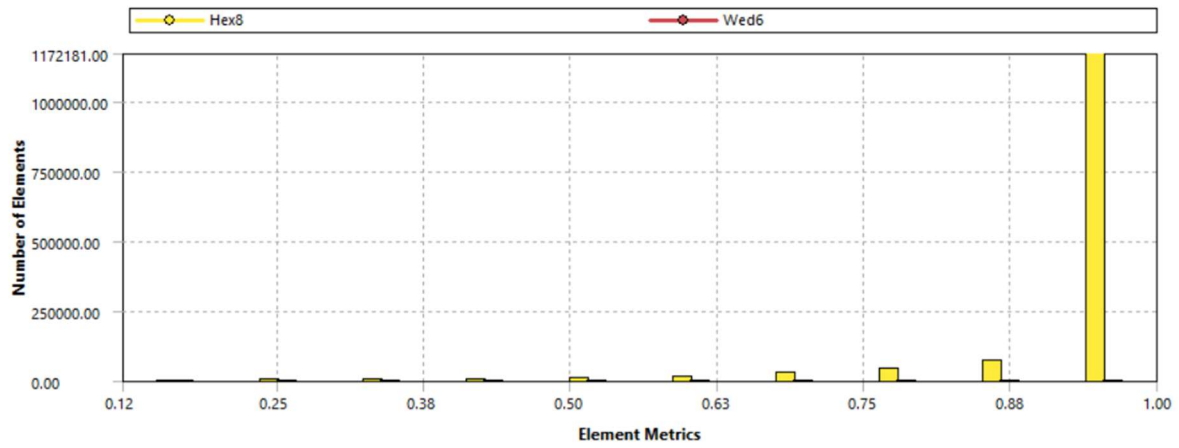


Figure 25: Orthogonal quality for curve 1

Mesh metrics for curve 2

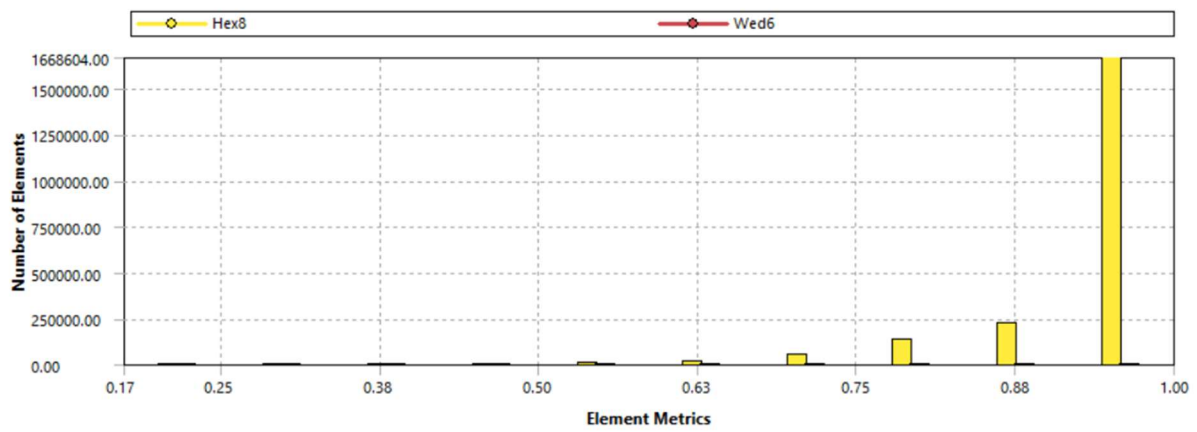


Figure 26: Element quality for curve 2

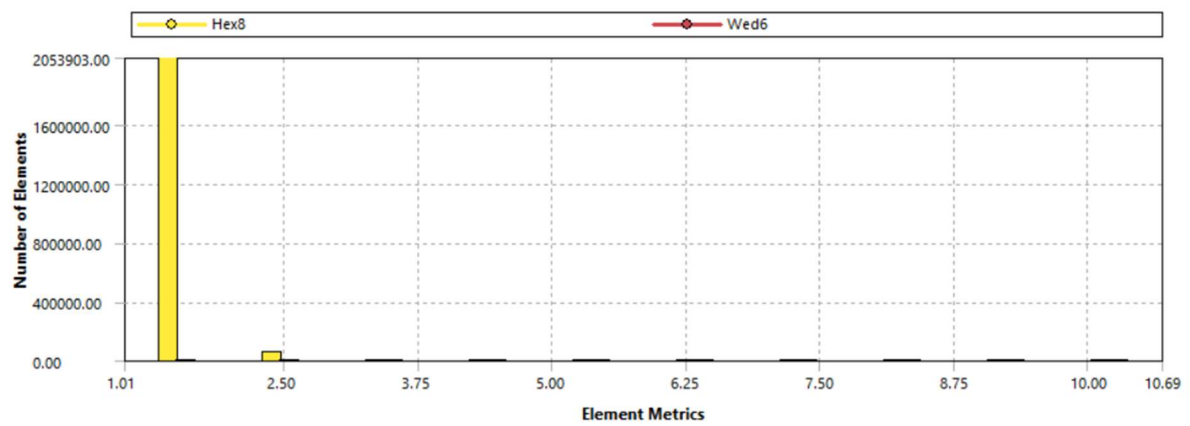


Figure 27: Aspect ratio for curve 2

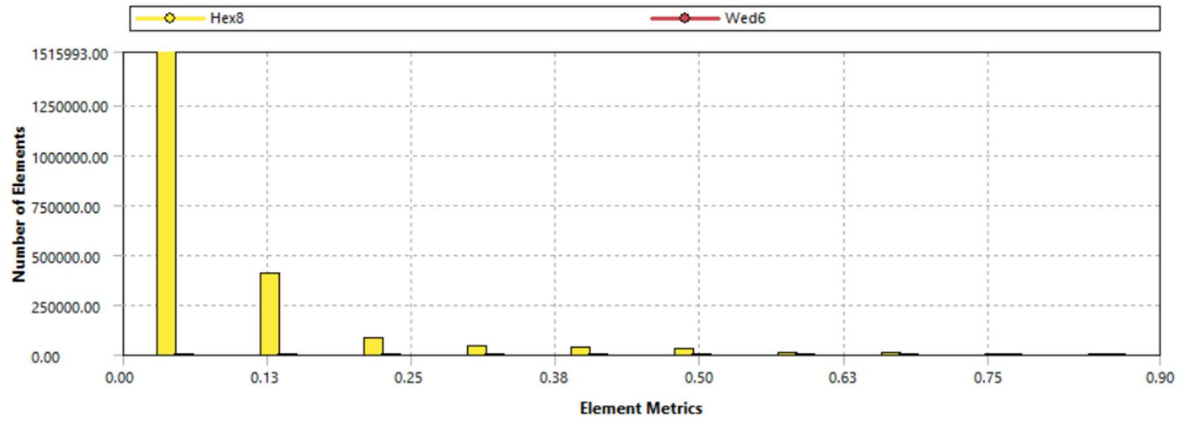


Figure 28: Skewness for curve 2

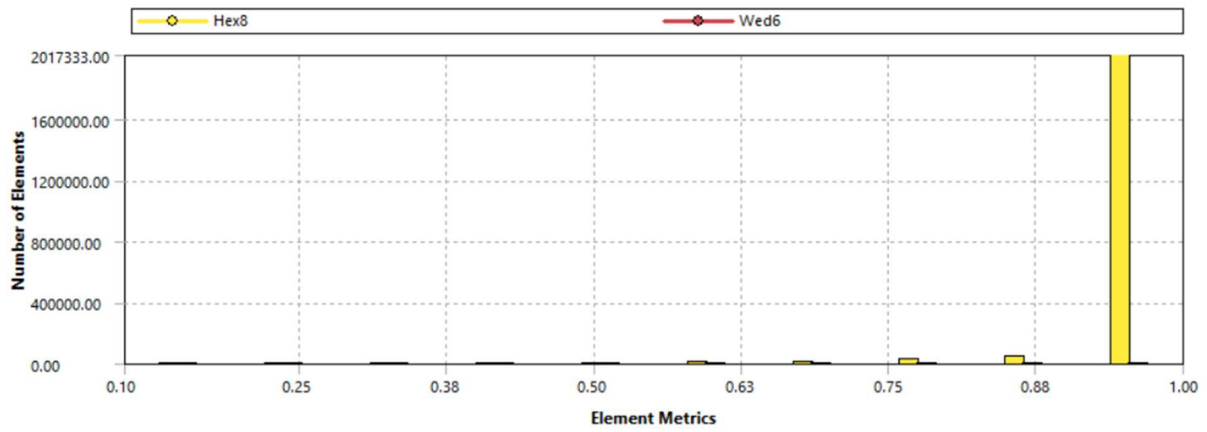


Figure 29: Orthogonal quality for curve 2

4 Simulation Settings

4.1 Convective flux calculation scheme

The Second Order Upwind Scheme is often ideal for applications with complex geometries and potentially turbulent flows such as a brake disk. QUICK scheme is commonly used in structured grids and is suitable for flows with strong directional bias and Low computational cost. Central Differencing Scheme is high in accuracy but can cause numerical instability in turbulent flows. A Taylor series expansion is used to compute convective fluxes at the cell faces using gradients, balancing accuracy with computational cost in this second order upwind. It is this scheme that captures gradients and flow details better than the First Order Upwind Scheme which can be overly diffusive in high gradient areas such as those present in brake disks. The automation of the modelling of the interactions between rotating disk and airflow is common in brake disk simulations, where the flow is usually turbulent. Second order schemes reduce numerical dissipation and hence improve the resolution of turbulence models. Simpler schemes handle the complicated features of the brake disk less effectively than the method [2].

4.2 Diffusive flux calculation scheme

The Central Differencing Scheme (CDS) is the most appropriate diffusive flux calculation scheme. Diffusion processes are smooth and continuous by nature. Second order accuracy central differencing scheme is also available and suits the diffusion dominated regions. The CDS uses the gradient of scalar quantities (such as velocity or species concentration) to calculate mass flow diffusion directly and achieves second order accuracy. This guarantees accurate mass diffusion resolution in the flow field [3].

4.3 Turbulence modelling

The flow conditions (Reynolds number, geometric complexity and the desired level of accuracy) are relevant for turbulence modelling. In the reference literature, however, the k omega method was employed. This is effective for high-speed turbulent flows and flow separation cases.

4.4 Pressure velocity coupling

The Coupled Scheme is the overall best one for this application. The Coupled scheme, by solving pressure and velocity simultaneously, improves accuracy, providing improved results in rotating and turbulent situations. High speed flows and high-pressure gradients are ideal for brake disk.

5 Results and Validation

5.1 Grid independency analysis

Due to the internal flow, the boundary is fixed. Inlet velocity and the outside pressure are constant according to the literature. Hence element size can be assumed as the parameter for the grid independency analysis. Mass flow rate with the parameter of mesh size is used to check the grid independency analysis.

P3 - Mesh Element Size ▾	P5 - Mesh Element Size ▾	P7 - Mesh Element Size ▾	P4 - mass_out-op ▾	P6 - mass_out-op ▾	P8 - mass_out-op ▾
mm	mm	mm	kg s ⁻¹	kg s ⁻¹	kg s ⁻¹
0.25	0.3	0.3	0	-0.0076336	-0.0069637
10	10	10	0	-0.0076303	-0.0069636
100	100	100	0	-0.0076855	-0.0069634
200	200	200	0	-0.0076063	-0.0069635

Figure 30: Mass flow rates with different mesh sizes

P3 – Curve 1

P5 – Curve 2

P7 - Baseline

P4 – Mass output in curve 1

P6 – Mass output in curve 2

P8 – Mass output in Baseline

According to the results mass flow rate is slightly varying with the element size.

5.2 Solution convergency study

Solution convergency can identify using scaled residuals vs iterations graphs for each simulation.

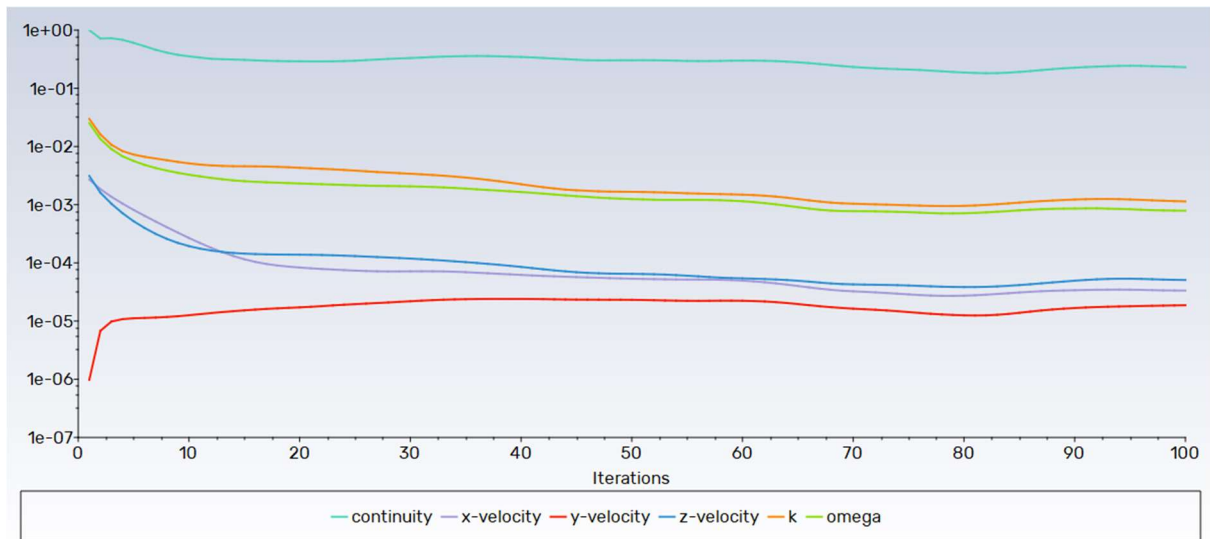


Figure 31: Scaled residuals vs iterations for baseline geometry

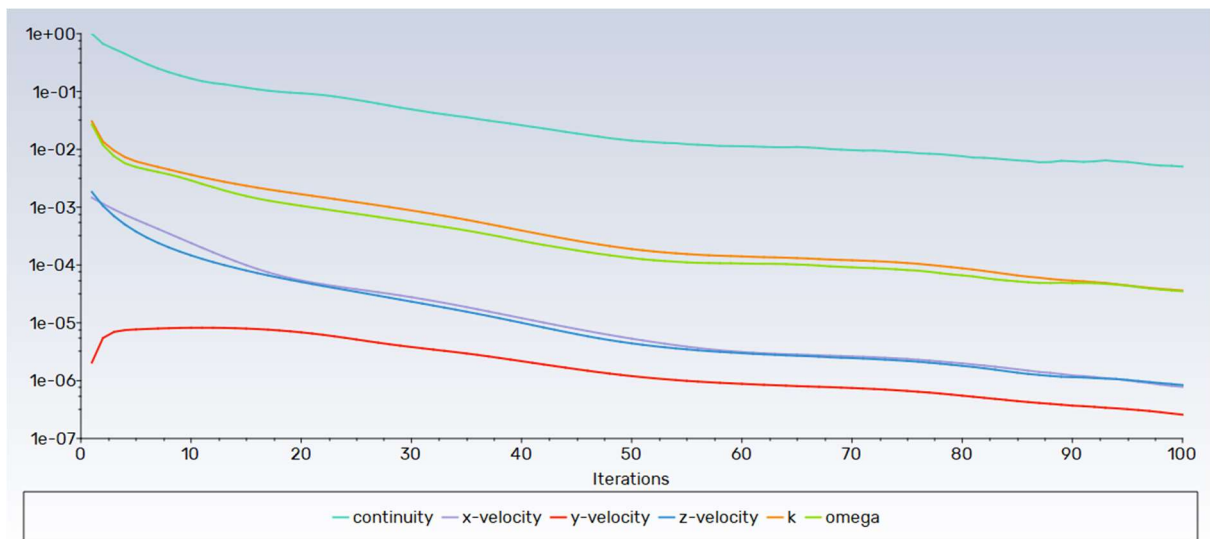


Figure 32: Scaled residuals vs iterations for geometry 1

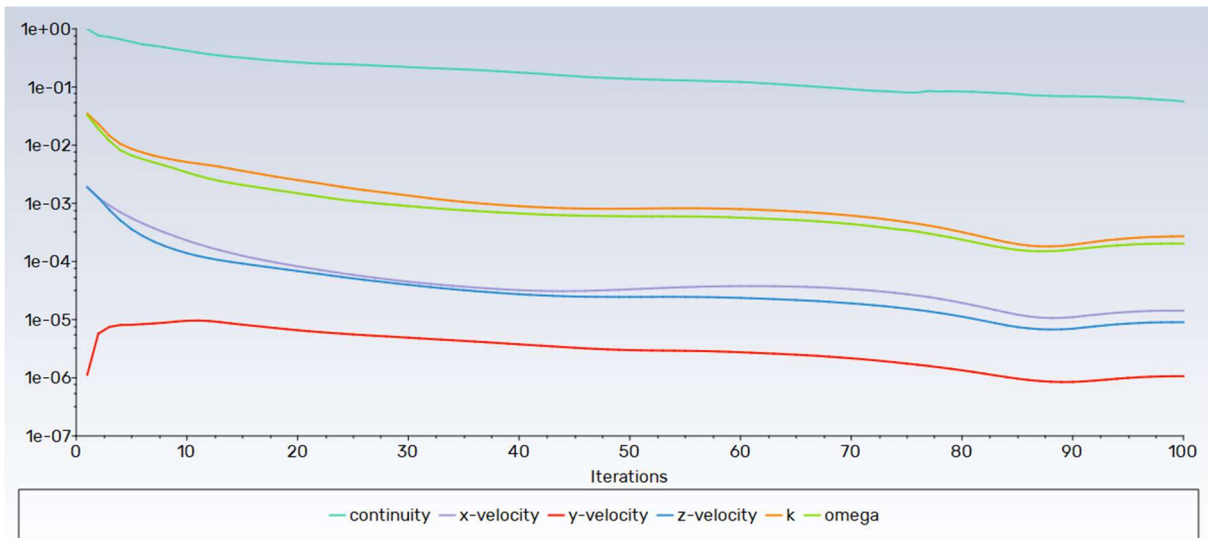


Figure 33: Scaled residuals vs iterations for geometry 2

5.3 Results

According to the literature, this experiment is conducted to remove the misalignment in basic disk brake system. So, they have plotted three velocity streamlines for each three types of the geometries.

The velocity streamline for the baseline:

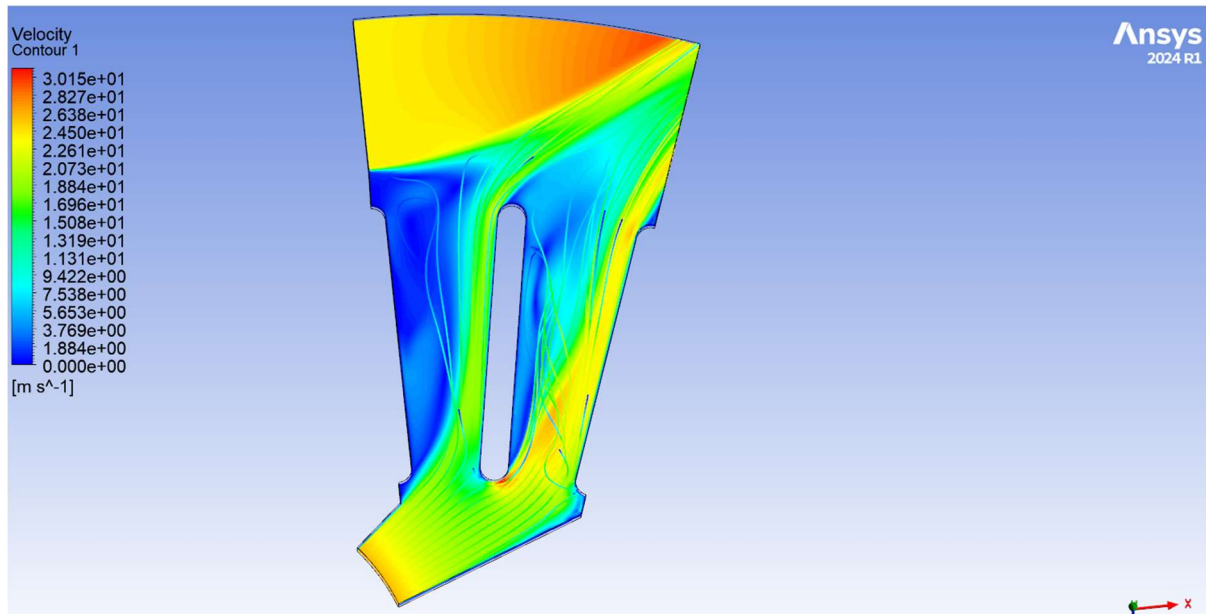


Figure 34: The velocity streamline for the baseline

The velocity streamline for the geometry 1:

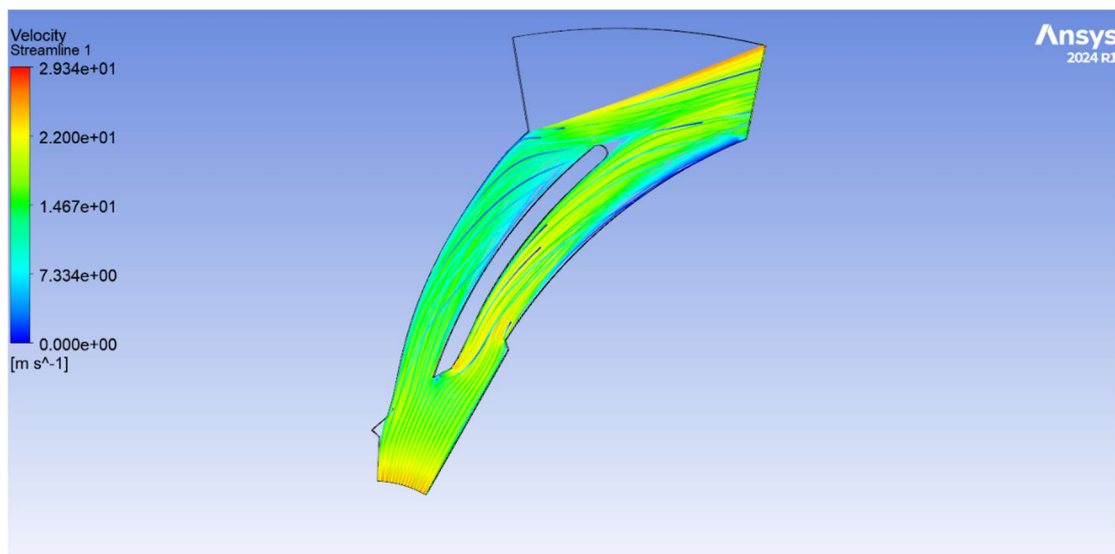


Figure 35: The velocity streamline for the curve 1

The velocity streamline for the geometry 2:

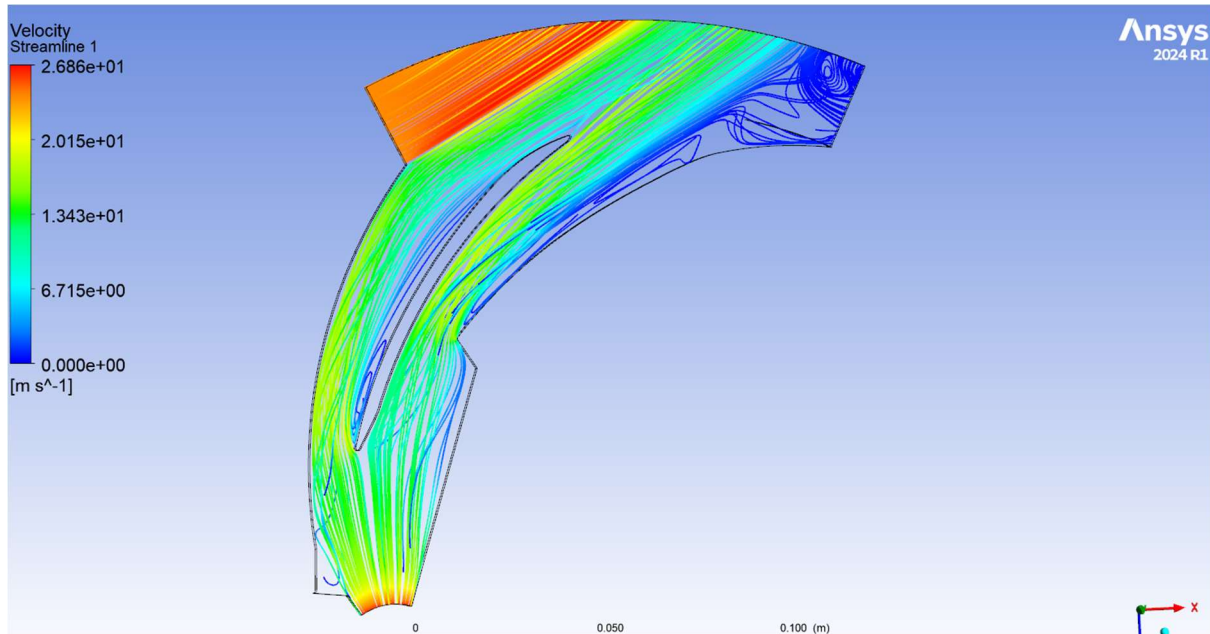


Figure 36: The velocity streamline for the geometry 2

According to the results, there is a flow misalignment in baseline geometry (figure 33). In two geometries, the flow misalignment is lesser in geometry 1 compared to the geometry 2.

5.4 Result validation

When comparing current simulation results with the document results, velocity streamlines are bit similar. But the magnitude of the maximum velocity is different.

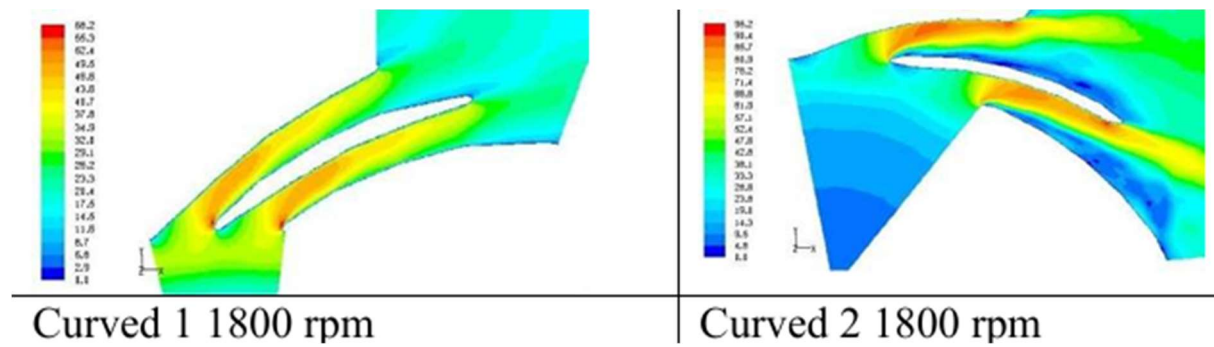


Figure 37: Research paper results of velocity streamlines for two geometries

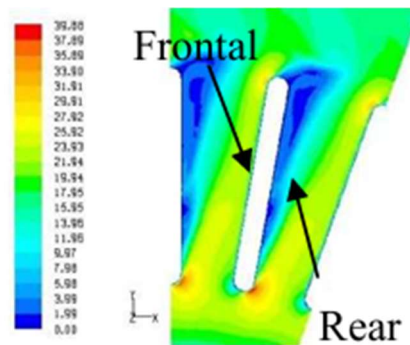


Figure 38: Research paper results of velocity streamlines for baseline

Table 3: Summary of the difference

Geometry	Maximum velocity (Current results)	Maximum velocity (Document results)	Difference
Baseline	30.15	39.48	medium
Geometry 1	29.34	56.2	high
Geometry 2	26.86	96.2	high

6 Discussion

6.1 Accuracy of the results

When it comes to the maximum velocity of the simulations, there is a huge difference between the current and literature results in both two new geometries. But in baseline geometry, it is around 25%.

But in the shape of velocity streamlines, streamlines are almost same as the current and literature results. Hence, overall results do not give good accuracy.

6.2 Possible errors in the simulation and what can be done to reduce the errors

Assumptions can be a reason for the errors. The gage pressure of the inside of a brake disk is not zero. But in this case, assumed that inside and the outside gage pressures are zero. And heat transfer also should be considered. But it also neglected.

A second possibility of error may come down to the mesh resolution being insufficient. According to the referenced research paper, medium mesh resolution with 2.5million elements was used. But in my simulations, I have taken the mesh resolution to be a 0.25mm, 3mm and 3mm element size with less than 2.5 million elements.

There have been convergence issues. Although the system stabilizes the residuals doesn't converge. These numerical errors can arise since stresses these directly coupled with stresses these.

6.3 Difficulties faced

During the simulation process, several challenges were encountered that impact the overall workflow, convergence and the accuracy of the results these difficulties primarily stemmed from the numerical setup, computational requirements and physical modelling complexities.

Most of the required data are not given in the literature. Inside and outside gage pressures are assumed to zero. Any data about heat transfer is not mentioned. The boundary data are not given. It is mentioned that mass flow rates were calculated using CFD, but the method is not mentioned. So, many assumptions are used, and many simulations are performed to get the almost same velocity streamlines as in the document.

Another challenge was to create an optimal mesh that is accurate, but also efficient computationally. High gradients were seen in areas like the outer regions of the vanes and near wall zones, requiring a high level of mesh refinement, which increased both the time and memory required to run the CFD.

6.4 What can be learned from the simulation results?

The results of the simulation provide a clear picture of airflow around the brake disk, including velocity distribution patterns. The importance of mesh resolution, and appropriate discretization schemes for capturing vortices and pressure gradients in the flow is also highlighted. Nevertheless, solver parameter management turned out to be a critical issue. It's about balancing accuracy with speed of computation. Improvements in the overall performance of the simulation were driven by insights from the grid independence study and solver adjustments.

7 Suggestions for Future Improvement

Assign group project with few members and give them to use more methods in CFD. For example, experiments with heat transfer, velocity contours, dynamic meshing, transient systems...etc. from a single group project.

References

- [1] Munisamy, Kannan M., et al. "Ventilated Brake Disk Air Streamlining Using Curved Vane." *Applied Mechanics and Materials*, vol. 225, Nov. 2012, pp. 26–31, <https://doi.org/10.4028/www.scientific.net/amm.225.26>.
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